- Exercise A

START 1	2	PIT 3
4	5	6
7	8	GOAL 9

1. Let's consider the generic Markov Decision Process (MPD) $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$, in the full Reinforcement Learning scenario (with $\mathcal{P}$ and $\mathcal{R}$ unknown). Provide the pseudo-code for the Monte Carlo prediction algorithm to estimate V_π, for a generic policy π, starting from the availability of trajectories (or the capability to generate trajectories); report both the 'first-visit' (FV) and 'every-visit' (EV) version of the algorithm.
2. Consider the deterministic small gridworld reported in the figure above: the state space is $\mathcal{S} = \{1, \ldots, 9\}$ and the terminal states are reported in grey. Episodes always starts in START. The action space is $\mathcal{A} = \{\text{up}, \text{down}, \text{left}, \text{right}\}$ and the state transitions are the obvious ones. If an action takes the agent off the grid, the state remain unchanged. All actions are associated, no matter the state they are taken from, with a reward equals to -1, unless: (i) the agent arrives to the PIT where it takes -10; (ii) the agent arrives to the GOAL where it takes +10 (if you arrive to PIT or GOAL don't consider the -1 cost associated with the step transition). Consider a random policy π, ie:

$$\pi(\text{up}, s_j) = \pi(\text{left}, s_j) = \frac{1}{6}, \qquad \pi(\text{right}, s_j) = \pi(\text{down}, s_j) = \frac{1}{3}, \qquad \forall s_j \in \mathcal{S}$$

   Let's consider the following trajectory (s, a, r; s, a, ...) obtained by applying policy π:

   - Episode: s_1,left,-1; s_1,left,-1; s_1,left,-1; s_1,down,-1; s_4,right,-1; s_5,down,-1; s_8,left,-1; s_7,down,-1; s_7,right,-1; s_8,left,-1; s_7,down,-1; s_7,right,-1; s_8,left,-1; s_7,up,-1; s_4,down,-1; s_7,down,-1; s_7,right,-1; s_8,down,-1; s_8,down,-1; s_8,right,10.

   Exploiting the Monte Carlo prediction algorithm, estimate V_π by considering: (I) $\gamma = 1$; (II) initial estimation $V_\pi = (0, 0, -10, 0, 0, 0, 0, 0, 10)$. Report the procedure with calculations. Report both the FV and the EV version of the algorithm.
3. Report a Policy Improvement algorithm in the full Reinforcement Learning scenario.
4. Suppose now $\mathcal{P}$ is known. Based on the knowledge obtained at point 2, devise with Policy Improvement a new policy π'. Is π' an optimal policy?
5. Consider an off-policy scenario, where the behaviour policy b is as follow:

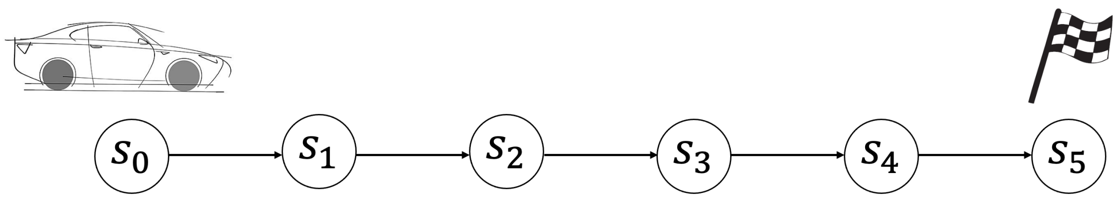
$$\underline{\ }(\text{up}, s_j) = \pi(\text{left}, s_j) = \pi(\text{down}, s_j) = \pi(\text{right}, s_j) = \frac{1}{4}, \qquad \forall s_j \in \mathcal{S}.$$

   Define the concept of coverage. Does b provides coverage for π (as defined at point 2)?
6. With b (as defined at point 5) the following trajectory (s, a, r; s, a, ...) has been obtained:

   - Episode: s_1,right,-1; s_2,right,-10.

   Estimate $Q_\pi(s_1, \text{right})$ and $Q_\pi(s_2, \text{right})$ based only on this episode. [You may consider a simplified version of the off-policy MC prediction algorithm where you have no previous estimate for Q_π and you rely only on the information obtained from this episode].

- Exercise B



Let's consider the generic Markov Decision Process (MPD) $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ and a generic policy π.

1. Provide the pseudo-code for the TD(0) algorithm to estimate V_π in the generic MDP; in the algorithm consider

- the incremental estimation for quantities evaluation based on the step size α.

2. Consider the 'driving home' example represented in the figure above: the goal of the agent is to estimate the remaining time before reaching the terminal state s_5 ('home'). The action space is $\mathcal{A} =$ drive and the state space is $\mathcal{S} = \{s_0, \ldots, s_5\}$. The state transitions are as depicted in the figure. Report the estimations for $V(s)$ with the Monte Carlo algorithm for policy evaluation considering:
   - a initial estimation $V(s_0) = 20$, $V(s_1) = 15$, $V(s_2) = 10$, $V(s_3) = 5$, $V(s_4) = 2$, $V(s_5) = 0$;
   - an episode where the collected rewards were (r_{t+1} is the notation for the reward associated with the transition from s_t and s_{t+1}): $r_1 = 8$, $r_2 = 4$, $r_3 = 6$, $r_4 = 3$, $r_5 = 2$;
   - $\alpha = 0.5$ and $\gamma = 1$.
3. Given the same settings described at the previous point, report the estimations for $V(s)$ with the TD(0) algorithm for policy evaluation.

REPORT YOUR ANSWERS TO EXERCISE A AND EXERCISE B IN THE PROTOCOL SHEET

- Question 1

Answer the following three questions:

1. Write the pseudo-code of the Sarsa Algorithm for estimating $Q \approx q^*$.
2. How is the Sarsa Algorithm modified into the Episodic Semi-gradient Sarsa Algorithm for estimating $\hat{q} \approx q^*$ (describe the main new updating steps) ?
3. How is the Sarsa Algorithm modified into the Sarsa(λ) Algorithm? (describe the main new updating steps)?

REPORT YOUR ANSWER TO QUESTION 1 IN THE FOLLOWING AND IN THE NEXT PAGES